

Algebra PhD Exam, May 2014

Do any 7 of the 11 problems, and mark clearly which problems you want graded if you try more than 7. State clearly any theorems that you use.

1. Suppose $p_1, p_2, \dots, p_r$ are distinct primes, and $K = \mathbb{Q}(\sqrt{p_1}, \dots, \sqrt{p_r})$, find the Galois group of $K/\mathbb{Q}$. How many fields F are there such that $\mathbb{Q} \subseteq F \subseteq K$?
2. This problem has two parts.
 - (i) Show that if K/k and L/K are separable algebraic extensions, then so is L/k .
 - (ii) Give an example of normal extensions K/k , L/K such that L/k is not normal.
3. Suppose K is an algebraic extension field of k and $\bar{k}$ is an algebraic closure of k . Show that K/k is separable if and only if the ring $\bar{k} \otimes_k K$ has no nonzero nilpotent elements.
4. Suppose A is a commutative ring with identity and $S \subset A$ is a multiplicative set not containing 0.
 - (i) Show that there is a prime ideal $\mathfrak{p} \subset A$ such that $\mathfrak{p} \cap S = \emptyset$.
 - (ii) Use part (i) to show that the intersection of all prime ideals of A is the set of nilpotent elements.
5. Suppose k is a field and A is a finitely generated commutative k -algebra. Let $\bar{k}$ be an algebraic closure of k . Note that the Galois group $G = \text{Gal}(\bar{k}/k)$ acts on the set $\text{Hom}_k(A, \bar{k})$ of k -algebra homomorphisms $A \rightarrow \bar{k}$ by composition. Construct a bijection of the set of orbits of G on $\text{Hom}_k(A, \bar{k})$ with the set of maximal ideals in A .
6. Suppose A is an integral domain. Show that an ideal $I \subseteq A$ is invertible if and only if it is projective and finitely generated.
7. Suppose X is a nonempty set and $F(X)$ is the free group on X . Show that $F(X)$ is abelian if and only if X is a singleton.
8. This problem has two parts.
 - (i) Define what it means for a group to be nilpotent.
 - (ii) Show that if G is a finite nilpotent group, then any Sylow subgroup is normal.
9. Suppose A is a commutative ring with identity and B is a commutative A -algebra with structure map $f : A \rightarrow B$. Show that the forgetful functor from the category of B -modules to the category of A -modules induced by f has a left adjoint.
10. Let p be a prime. Find all the isomorphism classes of noncommutative semisimple rings A of cardinality p^{10} .
11. Suppose R is a finite commutative ring with identity, such that $x^3 = x$ for all $x \in R$. Show that R is a finite direct sum of fields isomorphic to $\mathbb{F}_2$ or $\mathbb{F}_3$.